8.4.7 Prevention and Control

Integrated insecticidal spraying (DDT and Malathion) to reduce sandfly population.

Reduction of reservoir by killing all the infected dogs.

Personal prophylaxis by using anti – sandfly measures like using thick clothes, bed nets, window mesh or insect repellants and keeping the environment clean.

No vaccine is available against kala – azar.

8.5 Sporozoa – *Plasmodium*

Protozoan parasites characterised by the production of spore – like oocysts containing sporozoites were known as sporozoa. The parasites belonging to this group of protozoa do not possess any special organs of locomotion, such as flagella or cilia. The medically important parasite of this class that is given in the text is malaria parasites.

Malaria

It is the disease condition with seasonal intermittent fevers, chills and shivering. The name malaria (Mal: bad, aria: air) was given in the 18th century in Italy. The specific agent of malaria was discovered in RBC's of a patient in 1880 by Alphonse Laveran. In 1897, Ronald Ross identified the developing stages of malaria parasites in mosquitoes in Secunderabad, India. This led to various measures for the control and possible eradication of malaria by mosquito control. Both Ross (1902) and Laveran (1907) won the Nobel Prize for their discoveries in malaria.



The single most important protozoan disease is malaria, which causes 1.5 million deaths each year.

Different species of malaria parasites can develop in the same mosquito and such an infected mosquito can transmit the infection to man giving rise to cases of "mixed infection" the commonest being *P. falciparum* with *P. vivax*.

Infobits

Three basic types of malaria

1. Benign tertian (P. vivax and P. orale) with a fever every 2nd day (Example: Monday – fever, Tuesday – no fever, Wednesday – fever).
2. Benign quartan (P. malariae) with a fever every 3rd day. (Example: Monday; fever, Tuesday – no fever, Wednesday – no fever, Thursday – fever.
3. Malignant tertian (P. falciparum), in which the cold stage is less pronounced and the fever stage is more prolonged and severe. This type of malaria is more dangerous because of the complications caused by capillary blockage (i.e, convulsion, coma, acute pulmonary insufficiency and cardic failure). Large numbers of erythrocytes are parasitized and destroyed, which may result in dark-coloured urine. (black water fever); intravascular hemolysis, hemoglobinuria, and kidney failure).

   Two species of plasmodium, P. vivax and P. ovale, can remain in the liver, if not treated properly. The organism leave the liver and re-infect erythrocytes, causing the symptoms.

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